Lithium-Polymer Battery Capacity Prediction

Lithium-Polymer (LiPo) batteries are widely used in electric vehicles, energy storage systems, aerospace, and consumer electronics due to their high energy density and lightweight structure. Like all rechargeable batteries, they degrade over repeated charge–discharge cycles, reducing capacity and overall performance. This project uses real experimental data from a publicly available [Kaggle dataset]() containing cycle-wise capacity measurements and voltage/charge data collected under both Standard and Stress load conditions. Stress testing accelerates aging through higher current and thermal load, revealing differences in electrochemical behavior as the batteries deteriorate.

Traditional battery health estimation often relies on cycle count or expensive diagnostic equipment, neither of which captures true electrochemical aging in real time. By analyzing voltage and charge behavior with machine learning, subtle early indicators of degradation can be detected and used to more accurately predict capacity fade. The goal of this project is to develop a data-driven model that uses these electrical features to estimate LiPo battery health and forecast long-term performance.

Task 1 - Data Exploration

The analysis begins by reviewing the three datasets provided:

- **cycle_vs_capacity.csv**, containing measured battery capacity (Ah) for each charge–discharge cycle,
- **cycle_voltage_extracted_charge_std.csv**, recording high-frequency voltage and extracted charge under Standard operating conditions, and
- **cycle_voltage_extracted_charge_stress.csv**, capturing equivalent measurements under Stress loading. Stress testing simulates aggressive real-world conditions such as higher discharge current and thermal strain, which accelerates aging and exposes performance deterioration more clearly.

Merging the Datasets

Because capacity is measured once per cycle, while voltage and extracted charge are recorded many times within each cycle, the Standard and Stress datasets were aggregated by computing summary statistics (mean, min, max) for each battery and cycle. This transforms high-resolution sensor signals into cycle-level features that align with capacity values. After aggregation, the datasets were merged on battery_id and cycle_number, producing a unified table containing capacity and corresponding electrical metrics for each cycle.

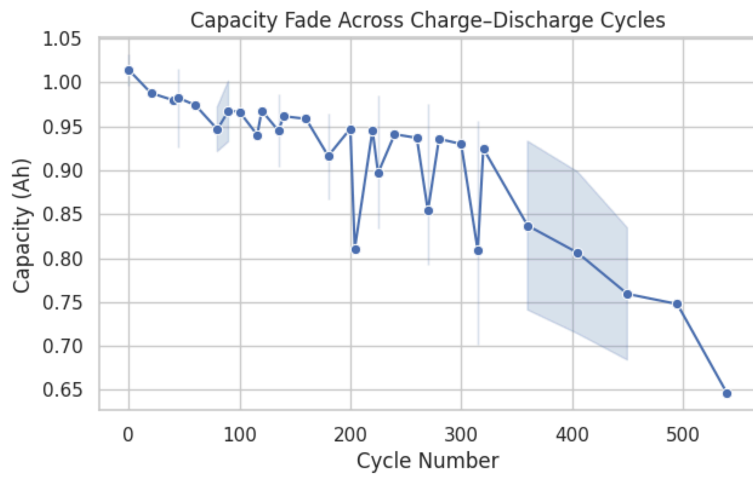
Handling Missing Values

The merged dataset revealed a small number of missing values, occurring only in the final recorded cycles for certain batteries where stress or standard measurements were not captured. To preserve chronological degradation trends and avoid removing meaningful capacity endpoints, linear interpolation was used to fill missing values. This method is appropriate because voltage and charge curve progression is naturally smooth across cycles, unlike mean/median imputation which could introduce bias or unrealistic discontinuities.

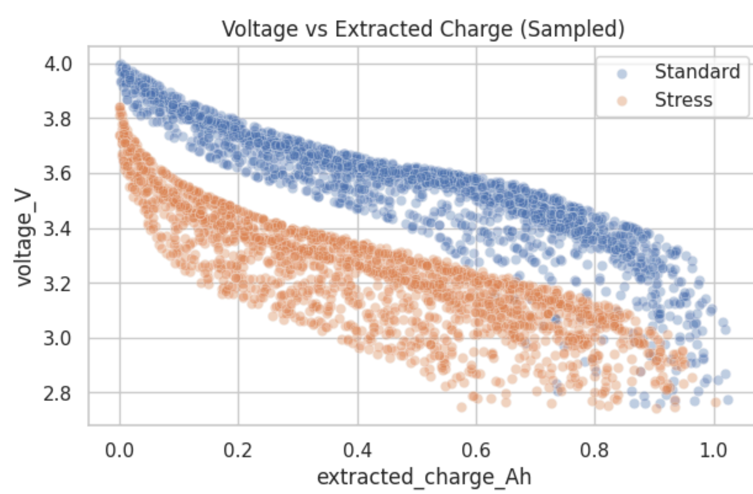
Visual Exploration

After merging and cleaning the dataset, exploratory analysis was performed to understand degradation behavior:

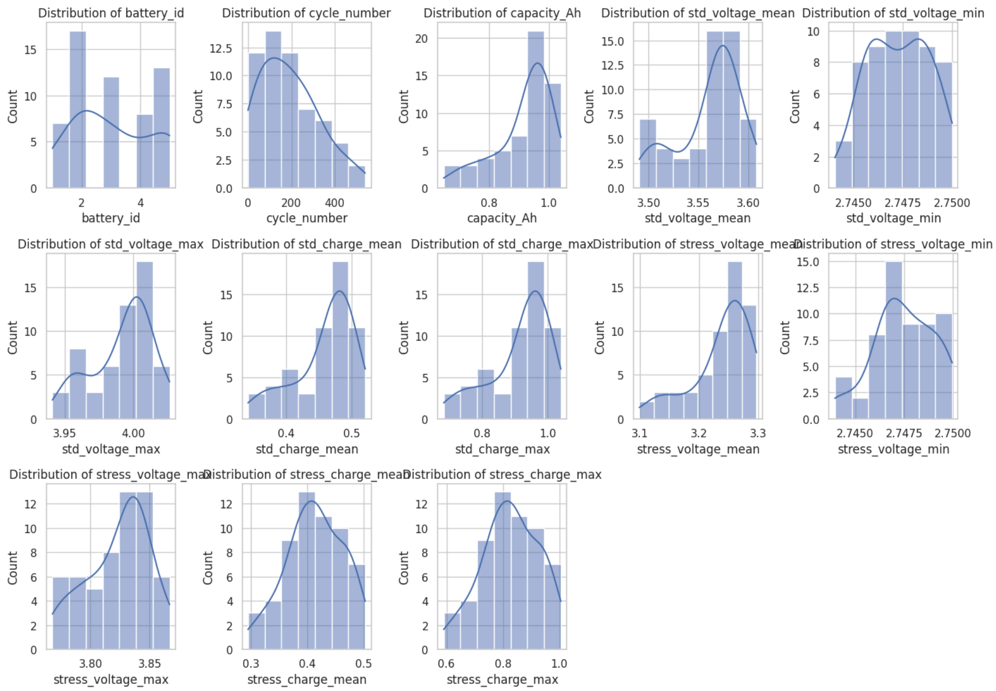
- **Capacity vs. Cycle Number** line plots show a predictable downward trend for all five batteries, confirming progressive aging.



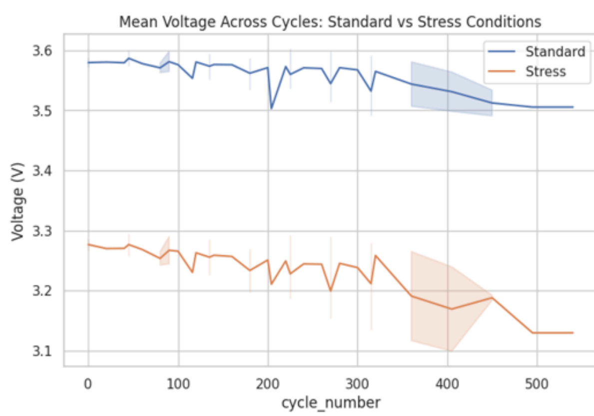
- **Voltage vs. Extracted Charge Scatter Plots** demonstrate steeper voltage decline under Stress load, indicating increased internal resistance and accelerated deterioration.

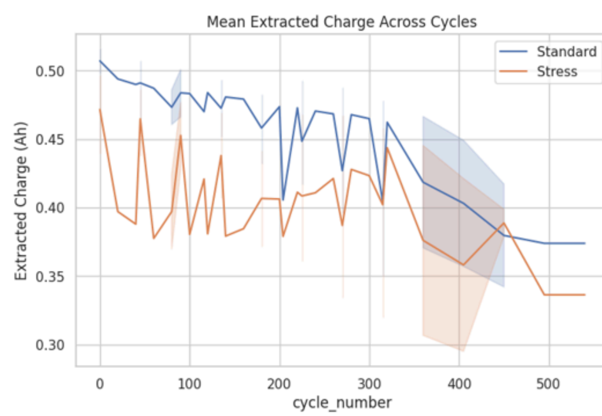


- **Distribution histograms** revealed smooth, well-behaved distributions without extreme outliers, suggesting consistent test conditions.



- **Trend plots of mean voltage and extracted charge** highlight increasing separation between Standard and Stress performance as cycles increase.





These insights validated assumptions about LiPo degradation and informed the subsequent feature engineering and modeling steps.

Task 2 - Feature Engineering

Following exploratory analysis, the next objective was to engineer meaningful predictive features that strengthen model performance while improving interpretability. Since the merged dataset contains multiple voltage and charge metrics recorded under both Standard and Stress testing conditions, it is important to evaluate redundancy and potential multicollinearity before building predictive models.

Multicollinearity Analysis

To examine feature redundancy, we calculated the **Variance Inflation Factor (VIF)** for all numerical predictors derived from the Standard and Stress performance metrics. VIF values significantly greater than 10 indicate serious multicollinearity, meaning the predictors carry nearly overlapping information and can destabilize regression-based models. The results showed extremely high VIF values across voltage and charge features (ranging from 10^6 to 10^{12}), which is expected because these measurements reflect the same electrochemical discharge curve behavior and naturally change together across battery cycles.

These results highlight that directly feeding all raw voltage and charge fields into a linear model would lead to unstable coefficients and poor interpretability. Although tree-based models like Random Forest and XGBoost can tolerate high collinearity, simplifying the feature set benefits generalization and computational efficiency.

Feature Engineering

To address redundancy and capture more meaningful physical behavior, we created delta features that quantify the deterioration gap between Standard and Stress performance:

$$merged\_df["voltage\_delta\_mean"] = merged\_df["std\_voltage\_mean"] - merged\_df["stress\_voltage\_mean"]$$
$$merged\_df["charge\_delta\_mean"] = merged\_df["std\_charge\_mean"] - merged\_df["stress\_charge\_mean"]$$

Why delta features matter

- Stress testing accelerates aging, so increasing performance separation reflects internal resistance growth and capacity fade.
- Voltage drops more rapidly under stress when a battery deteriorates; therefore, the difference between standard and stress voltage is a strong early degradation indicator.
- These features significantly reduced redundancy while providing physically interpretable signals relevant to battery health monitoring.

Feature Selection for Modeling

After engineering new features and reviewing updated correlation heatmaps, we selected a reduced set of predictors such as cycle_number, voltage_delta_mean and charge_delta_mean because:

- cycle_number captures natural chronological degradation,
- voltage_delta_mean proved to be the strongest degradation indicator,
- charge_delta_mean provides secondary information about usable energy reduction.

Voltage_mean and charge_mean features were removed due to high multicollinearity and diminishing marginal benefit.

Task 3 - Modeling

The objective of this stage is to build a machine learning model capable of predicting battery capacity (Ah) as it degrades over repeated charge–discharge cycles. Since the target variable is continuous, this is a regression problem.

Model Selection

To evaluate different perspectives of the degradation pattern, three regression models were trained and compared:

Model	Reason for Selection
Linear Regression	Baseline model to capture linear degradation relationship with cycle count and engineered features. Easy to interpret.
Random Forest Regressor	Handles non-linear behavior, resistant to multicollinearity, good performance on small tabular datasets.
XGBoost Regressor	Gradient boosting models are highly effective at modeling complex nonlinear degradation patterns and feature interactions.

The dataset was split into 80% training and 20% testing to evaluate model generalization.

Model Performance

Performance was assessed using Root Mean Squared Error (RMSE) and R^2 Score, which reflect prediction accuracy and variance explained.

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Linear Regression → RMSE: 0.0395, R²: 0.8517

Random Forest     → RMSE: 0.0230, R²: 0.9495

XGBoost           → RMSE: 0.0306, R²: 0.9107
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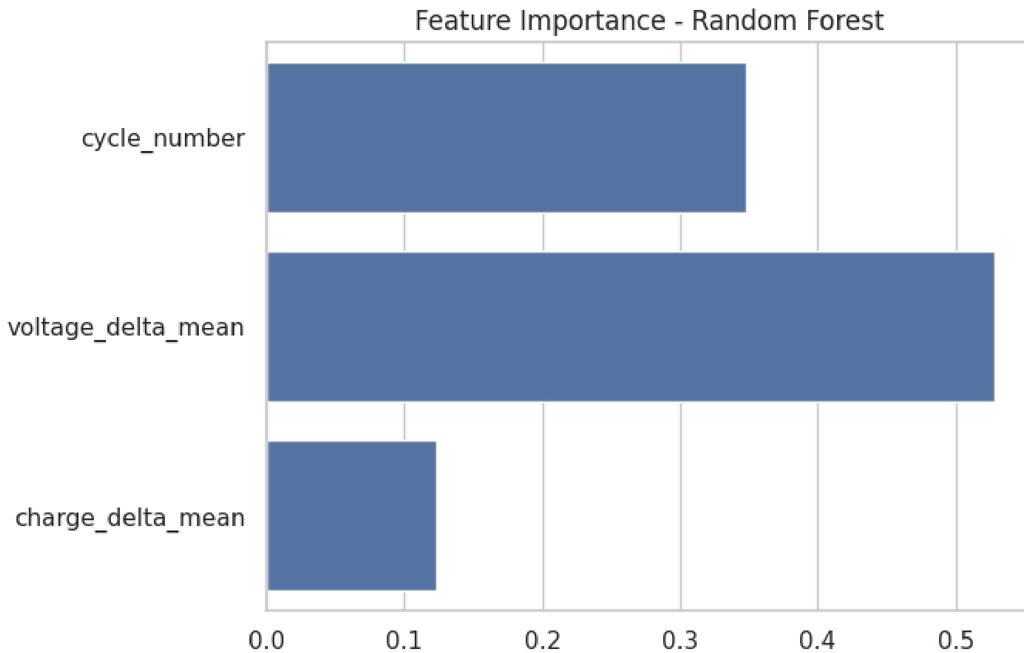
Random Forest produced the best results, achieving the lowest RMSE and highest R^2. This indicates that battery capacity degradation exhibits non-linear behavior influenced by interaction between cycle count and electrochemical performance differences under load.

Feature Importance Analysis (Random Forest)

Feature importance scores highlight the relative contribution of each engineered feature:

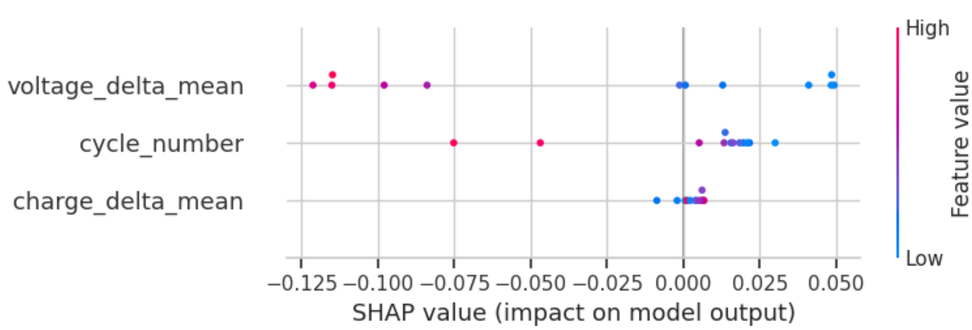


Key observations:

- voltage_delta_mean is the most influential feature, demonstrating that the deterioration gap between Standard and Stress voltage performance is the strongest indicator of internal resistance increase and electrochemical aging.
- cycle_number remains an important predictor, confirming expected gradual degradation with usage.
- charge_delta_mean has lower importance, indicating that voltage changes reflect degradation earlier and more reliably than charge measurements.

Explainability via SHAP Values

SHAP analysis offers deeper interpretability, showing how features influence individual predictions.



Insights from SHAP:

- High values of voltage_delta_mean strongly reduce predicted capacity, signaling rapid deterioration.
- Lower cycle numbers correspond to higher predicted capacity (i.e., early lifecycle health).
- Model behavior aligns closely with real electrochemical behavior, validating trust in predictions.

Summary of Key Modeling Insights

- Battery degradation is nonlinear and best modeled using tree-based ML models.
- The voltage performance gap under stress is the earliest and strongest indicator of capacity fade.
- Cycle count alone is insufficient as an estimator incorporating electrical behavior dramatically improves prediction performance.
- The Random Forest model effectively captures degradation patterns, achieving $R^2 \approx$ 95%